work done on the gas which turns out to be the area with a negative sign.

Note

To calculate the work done in an isothermal process, we assume that the process is quasi-static. If it is not quasi-static we cannot substitute $P = \frac{\mu RT}{V}$ in equation (8.31). It is because the ideal gas law is not valid for non equilibrium states. But equation (8.34) is valid even when the isothermal process is not quasi-static. This is because the state variables like pressure and volume depend on initial and final state alone of an ideal gas and do not depend on the way the final state is reached. The assumption of 'quasi-static' requires to do the integration.

EXAMPLE 8.16

A 0.5 mole of gas at temperature 300 K expands isothermally from an initial volume of 2 L to 6 L

(a) What is the work done by the gas?

(b) Estimate the heat added to the gas?

(c) What is the final pressure of the gas?

(The value of gas constant, R = 8.31 J mol^{-1} K^{-1})

Solution

(a) We know that work done by the gas in an isothermal expansion

Since μ = 0.5

$$W = 0.5mol \times \frac{8.31J}{mol.K} \times 300\ K\ \text{In}\left(\frac{6L}{2L}\right)$$

$$W = 1.369\ kJ$$

Note that W is positive since the work is done by the gas.

(b) From the First law of thermodynamics, in an isothermal process the heat supplied is spent to do work.

Therefore, $Q = W = 1.369\ kJ$. Thus Q is also positive which implies that heat flows in to the system.

(c) For an isothermal process

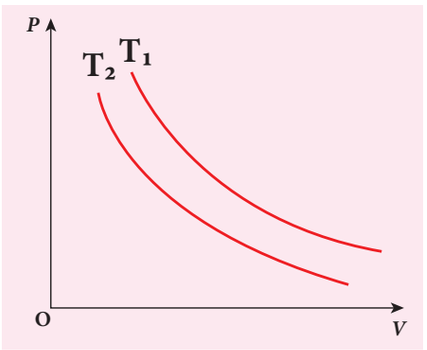
$$P_i V_i = P_f V_f = \mu RT$$

$$P_f = \frac{\mu RT}{V_f} = 0.5\text{mol} \times \frac{8.31J}{mol.\text{K}} \times \frac{300K}{6\times10^{-3}m^3}$$

$$=207.75\ k\,Pa$$

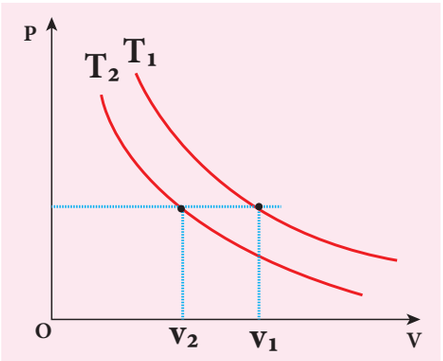
EXAMPLE 8.17

The following PV curve shows two isothermal processes for two different temperatures and. Identify the higher temperature of these two.



Solution

To determine the curve corresponding to higher temperature, draw a horizontal line parallel to x axis as shown in the figure. This is the constant pressure line. The volumes V_1 and V_2 belong to same pressure as the vertical lines from V_1 and V_2 meet the constant pressure line.



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